

# Introduction to QFT notes

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Prof. Boglione, notes by Paolo Rocchietti March<sup>1</sup>

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<sup>1</sup>paolo.rocchiettimarc@edu.unito.it

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# 1 Lesson 1, 23/09

## PRIMA LEZIONE SU TABLET

# 2 Lesson 2, 24/09

## 2.1 Natural units

From this point onwards, natural units will be used:

$$\hbar = c = 1$$

**Note:** this messes up dimensionality, but, conveniently, forces mass and energy to have the same units

$$E = mc^2 \xrightarrow{c=1} E = m,$$

and length and time to have the same dimensionality

$$x = ct \xrightarrow{c=1} [x] = [t].$$

To convert back into more familiar units the constants are, respectively,

$$c = 299.792.458 \text{ m/s}$$

$$\hbar c \approx 0.197 \text{ GeV}\cdot\text{fm}$$

## 2.2 Klein-Gordon Equation

The relativistic quantum mechanics' fundamental equation is

$$(\square + m^2)\psi(x) = 0$$

where

$$\square = \partial^\mu \partial_\mu = \frac{\partial^2}{\partial t^2} - \nabla^2$$

and  $\psi(x)$  is still the Schrödinger's wavefunction.

It will now be shown that this equation **does not** conserve probability, using a continuity equation.

Regular quantum mechanics states

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{j} = 0,$$

where

$$\rho = \langle \psi_t | \psi_t \rangle \quad \mathbf{j} = \frac{i}{m} (\psi^* \nabla \psi - \psi \nabla \psi^*)$$

Consider the Klein-Gordon equation left-multiplied by  $\psi^*(x)$  and its conjugate

$$\begin{cases} \psi^*(x) (\square + m^2) \psi(x) = 0 \\ \psi(x) (\square + m^2) \psi^*(x) = 0 \end{cases}$$

their difference is

$$\begin{aligned} \psi^*(\square + m^2)\psi - \psi(\square + m^2)\psi^* &= 0 \\ \partial^\mu (\psi^* \partial_\mu \psi - \psi \partial_\mu \psi^*) &= 0 \\ \partial^\mu J_\mu &= 0 \end{aligned}$$

this is a 4D continuity equation and  $J^\mu$  is a Noether current.

Expanding the equation yields

$$\frac{\partial}{\partial t} \left( \psi^* \frac{\partial \psi}{\partial t} - \psi \frac{\partial \psi^*}{\partial t} \right) - \nabla \cdot (\psi^* \nabla \psi - \psi \nabla \psi^*) = 0$$

the second term is proportional to the quantum probability current, but it is missing an  $i$ ! Moreover, the first term, representing  $\rho$ , is **not definite positive**, so it cannot represent the norm of the state  $|\psi_t\rangle$ !

This proves that it is **impossible** to make QM relativistic using wavefunctions and retaining the probabilistic approach.

Paul Dirac used another approach, he wanted to decrease the order of space derivative (instead of increasing the time derivative order, like K-G): this result was better, but for free particles it yielded a double solution with  $E < 0$  states, violating a fundamental QM theorem!

For this reason, he devised the concept of the *Dirac sea*: these  $E < 0$  states must all be occupied (because they cannot be observed), and a uniform charge can be factored out of Maxwell's equations with proper boundary conditions; when one of these particles in the sea is excited, the "hole" left behind can be perceived as a "positive electron": he predicted the existence of *antiparticles*.

This interpretation is still classical, but it begins to convey the concept that a finite number of particles is *not* possible for relativistic QM, so the concept of wavefunctions must be left behind in favour of field theories.

## 2.3 Symmetries and conservations

To build a theory, it is necessary to start with principles and symmetry: **Lorentz-invariance** to guarantee the conservation of distance between points (this is also useful to define a metric topology!).

Why symmetry? Noether theorem states that for every 4-current that respects a continuity equation

$$\partial_\mu J^\mu = 0$$

there is a conserved Noether charge

$$Q := \int d^3x J^0, \text{ so that } \frac{dQ}{dt} = 0$$

### 2.3.1 Lorentz-Poincaré group

The Lorentz invariant is

$$s^2 = t^2 - x_i x^i = t'^2 - x'_i x'^i = x_\mu x^\mu :$$

the Lorentz group is the set of all transformations  $x^\mu \rightarrow x'^\mu$  over the Minkowski space that conserve  $s^2$ ; in matrix representation the elements are

$$x'^\mu = \Lambda^\mu_\nu x^\nu$$

the invariant is conserved **if and only if**

$$g_{\mu\nu} \Lambda^\mu_\rho \Lambda^\nu_\sigma = g_{\rho\sigma}.$$

The covariant quantities transform with the inverse matrix

$$x'_\mu = \Lambda_\nu^\mu x_\nu$$

**Observation 1** Since  $g_{\mu\nu}$  is a symmetric matrix, this equation encompasses 10 constraints on  $\Lambda^\mu_\nu$ , leaving 3 degrees of freedom for rotations and 3 for boosts.

**Naming convention** Consider the determinant of the invariance condition:

$$\det(\Lambda^T g \Lambda) = \det(g) \leftrightarrow \det(\Lambda)^2 = 1 \leftrightarrow \det(\Lambda) = \pm 1$$

Transformations with  $\det(\Lambda) = 1$  are called *proper Lorentz transformations*.

Transformations with  $\det(\Lambda) = -1$  are called *improper Lorentz transformations*.

**Naming convention** Consider the (0,0) component of the invariance condition:

$$g_{\mu\nu} \Lambda^\mu{}_0 \Lambda^\nu{}_0 = 1 \leftrightarrow (\Lambda_{00})^2 - \Lambda^i{}_{0g} \Lambda^i{}_0 = 1 \leftrightarrow |\Lambda^0{}_0|^2 \geq 1$$

Because this term regulates how the transformation relates  $x^0$  and  $x'^0$ , transformations with  $\Lambda^0{}_0 \geq 1$  are called *orthochronous Lorentz transformations*, while those with  $\Lambda^0{}_0 \leq -1$  are called *non-orthochronous Lorentz transformations*.

These 2 divisions split the Lorentz group into 4 parts

- Proper and orthochronous  $L_+^\uparrow$ : restricted Lorentz group, sometimes referred to as "Lorentz Group";
- Improper and orthochronous  $L_-^\uparrow$ : spacial inversion;
- Proper and non-orthochronous  $L_+^\downarrow$ : time inversion;
- Improper and non-orthochronous  $L_-^\downarrow$ : total inversion of spacetime.

**Note** The only subgroup is  $L_+^\uparrow$ , as it is the only one that contains the identity matrix  $\mathbb{I}$ .

## 3 Lesson 3, 28/09

### 3.1 Symmetries and Group Theory

The principle of symmetry (invariance) is the guiding light to create a theory: the first and fundamental symmetry is *Lorentz invariance*, because special relativity states that **all physical laws must be the same in every inertial reference frame**; each particle (state) must also observe other symmetries, but those vary case by case.

The language of symmetries is *group theory*, so, in order to proceed, an introduction to groups is needed.

**Def.** A group is a set  $G$  with a binary operation  $\cdot$  so that

- $\forall g_1, g_2 \in G, g_1 \cdot g_2 \in G$ ;
- $\exists e \in G$  so that  $\forall g \in G, g \cdot e = e \cdot g = g$ ;
- $\forall g \in G, \exists g^{-1}$ , so that  $g^{-1} \cdot g = g \cdot g^{-1} = e$ ;
- $\forall g_1, g_2, g_3 \in G, g_1 \cdot (g_2 \cdot g_3) = (g_1 \cdot g_2) \cdot g_3 = g_1 \cdot g_2 \cdot g_3$ .

**Def.** A group is *Abelian* if and only if

$$\forall g_1, g_2 \in G, g_1 \cdot g_2 = g_2 \cdot g_1$$

#### 3.1.1 Zoology

- $U$  meaning "unitary" means  $UU^\dagger = \mathbb{I}$ , for complex valued matrices;
- $O$  meaning "orthogonal" means  $OO^T = \mathbb{I}$ , for real valued matrices
- $S$  meaning "special" means  $\det(S) = 1$ .

## Examples

- $U(1)$  is the group of phase rotations in  $\mathbb{C}$ ;
- $SU(2)$  is the group of spin;
- $SO(3)$  is the group of proper rotations

**Note** The number in the name of the group is the dimensionality of the fundamental representation.

## 3.2 Lie Groups

The most important groups are those that represent *continuous transformations*, i.e. every element of the group corresponds to a transformation that can be broken down into *infinitesimal transformations*: using parameters  $\alpha$  and generators  $T$ , the element  $g(\alpha) \in G$  is

$$g(\alpha) = \mathbb{I} + i\alpha_a T^a + O(\alpha^2) = \dots = \exp[i\alpha_a T^a]$$

so its generators can be found as

$$T^a = -i \left. \frac{dg}{d\alpha_a} \right|_{\alpha=0}$$

it is useful to work with generators as their number is small and they are operators (that can be represented as matrices).

The operation of the groups can be used to construct the operation between the generators, by the BCH formula:

$$g_1 \cdot g_2 \rightarrow e^X \cdot e^Y := \exp \left( X + Y + \frac{1}{2}[X, Y] + \frac{1}{12}[X, [X, Y]] \right)$$

where  $[\cdot, \cdot]$  is the *Lie commutator*.

**Def.** The vector space of the generators equipped with a commutator (bilinear, anti-commutative, Jacobi Identity) is called a **Lie Algebra**  $\mathfrak{g}$ .

**Note** The most general form of a commutator is

$$[T^a, T^b] = if_{bc}^a T^c$$

### 3.2.1 Example: algebra of $SO(3)$

The rotation matrix about the X axis of an angle  $\vartheta_1$  is

$$R_1(\vartheta_1) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos(\vartheta_1) & -\sin(\vartheta_1) \\ 0 & \sin(\vartheta_1) & \cos(\vartheta_1) \end{pmatrix}$$

the first generator is then

$$J_1 = -i \left. \frac{dR_1}{d\vartheta_1} \right|_{\vartheta_1=0} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix}.$$

The other 2 matrices can be obtained similarly:

$$J_2 = \begin{pmatrix} 0 & 0 & i \\ 0 & 0 & 0 \\ -i & 0 & 0 \end{pmatrix} \quad J_3 = \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

The commutation relations turn out to be

$$[J^i, J^j] = i\varepsilon_{ijk} J^k.$$

From the generators, the  $J^\pm$  operators are constructed.

### 3.2.2 Example: algebra of $SU(2)$

The definition states  $SU(2) = \{2 \times 2 \text{ unitary matrices with } \det(U) = 1\}$ : the conditions imply, by the BCH formula:

$$U^\dagger U = \mathbb{I} \rightarrow J_i^\dagger = J_i$$

$$\det(U) = \det(e^{J_i}) = 1 = \exp[\text{Tr}(J_i)] \rightarrow \text{Tr}(J_i) = 0$$

the traceless, hermitian matrix vector space has base  $\sigma^i$ , the Pauli matrices, which have commutation relations

$$[\sigma^i, \sigma^j] = i\varepsilon_{ijk}\sigma^k$$

### 3.3 Group representations

**Def.** A representation of a group  $G$  is a map

$$r : G \rightarrow GL(V)$$

$$r : g \rightarrow \text{square matrix}$$

so that

$$r(g_1 \cdot g_2) = r(g_1) \cdot r(g_2)$$

where the second  $\cdot$  is the matrix multiplication.

$\dim(V) = n$  is the dimension of the representation.

**Def.** An irreducible representation is a representation in which no subset of  $\text{Im}(r)$  is a subgroup of  $\text{Im}(r)$ .

**Note** A reducible representation can always be block-diagonalized (?).

## 4 Lesson 4, 30/09

The generators are useful because one can study the entire group just by considering the generators: they are operators uniquely defined by their commutation relations using Lie brackets; the set of generators equipped with their commutation relations is the **Lie algebra** associated with the group.

Once the algebra is known, the choice of the representation yields matrices equivalent to the generators.

As first examples, it is useful to consider  $SO(3)$  and  $SU(2)$ , because they are closely related to the Lorentz group (which is the final purpose of this parenthesis on group theory).

### 4.1 Example: angular momentum as a representation

As is known, the algebra of the angular momentum operators  $J^i$  is

$$[J^i, J^j] = i\varepsilon^{ij}_k J^k$$

the rank of a group is the maximum number of generators that commute with one another: in this case the rank is 1 and the chosen operator is  $J^3$ .

A Casimir operator is an operator that commutes with all generators: in this case it is

$$J^2 = J_i J^i.$$

The physical question is then "how many possible values/eigenstate of the angular momentum are possible?" and it turns out that the answer to this question also answers, in a QFT formalism, "how many particles does the field predict?".

The process to find the possible eigenvalues and eigenstates is known (see QM1 course) and it yields the

fact that for each  $j$  value, the number of eigenstates of  $J^3$  is finite and their eigenvalues  $m$  must be in the range

$$-j < m < j$$

where  $j$  is called the *maximum weight*; so for each  $j$  the possible values of  $J^3$  (and so the number of different eigenstates) are  $2j + 1$ . From this process also follows that the only possible values of  $j$  are positive integers or semi-integers and that the values of  $m$  always differ by an integer (so if  $j$  is semi-integer, all the corresponding values of  $m$  must also be semi-integer and vice-versa).

This formalism is very powerful because it relates the irreducible representations of the algebra to its vector basis: in a QFT, the dimension of the vector space will correspond to the number of possible physical particles.

This algebra is common between  $SU(2)$  and  $SO(3)$ , with the key difference that for  $SO(3)$  only integer values of  $j$  are permitted: as seen in intro groups,  $SU(2)$  can be represented as a 3-sphere immersed in  $\mathbb{R}^4$ , while  $SO(3)$  is only half 3-sphere in which antipodal points are mapped to the equator (??).

## 4.2 Example: reducible representation

The combination of 2 spin  $\frac{1}{2}$  particles yields a reducible representation: the vector space is

$$V = \mathbb{C}^2 \otimes \mathbb{C}^2, \quad \dim(V) = 4$$

again, the process of spin combination is treated in the QM1 and QM2 courses and it yields a *triplet* and a *singlet* of states:

$$\text{Triplet (J=1): } \begin{cases} |J=1, M=-1\rangle \\ |J=1, M=0\rangle \\ |J=1, M=1\rangle \end{cases} \rightarrow \text{Vector states}$$

$$\text{Singlet J=0 : } |J=0, M=0\rangle \rightarrow \text{Scalar state}$$

So the space  $V$  is subdivided into two different vector spaces which differ in symmetry: the singlet is antisymmetric while the triplet is symmetric; this representation is reducible, because the image of the representation of the algebra is the direct sum of the images of the representations of two different algebras.

## 4.3 The Lorentz Group Representation

Only the subgroup  $L_+^\uparrow$  is here considered and its formal name is  $SO(1,3)$ , where  $(1,3)$  is the signature of the Minkowski metric  $(+, -, -, -)$ .

Consider an infinitesimal Lorentz transformation

$$\Lambda^\mu{}_\nu = \delta^\mu{}_\nu + \omega^\mu{}_\nu, \quad |\omega^\mu{}_\nu| \ll 1 \quad \forall \mu, \nu$$

where  $\omega^\mu{}_\nu$  are called *parameters* of the transformation.

Substituting into the invariance condition yields

$$\begin{aligned} g_{\mu\nu} \Lambda^\mu{}_\rho \Lambda^\nu{}_\sigma &= g_{\rho\sigma} \\ g_{\mu\nu} (\delta^\mu{}_\rho + \omega^\mu{}_\rho) (\delta^\nu{}_\sigma + \omega^\nu{}_\sigma) &= g_{\rho\sigma} \\ g_{\mu\nu} \delta^\mu{}_\rho \delta^\nu{}_\sigma + \omega_{\nu\rho} \delta^\nu{}_\sigma + \omega_{\mu\sigma} \delta^\mu{}_\rho &= g_{\rho\sigma} \\ \omega_{\rho\sigma} &= -\omega_{\sigma\rho} \end{aligned}$$

so the parameters must be **antisymmetric**. This confirms the fact that each transformation is uniquely determined by 6 quantities. The transformation becomes

$$\Lambda^\mu{}_\nu = \delta^\mu{}_\nu + g^{\mu\rho} g^\sigma{}_\nu \omega_{\rho\sigma}$$

since  $\omega_{\rho\sigma}$  are antisymmetric, the only non-zero part of the product  $\omega_{\rho\sigma} g^{\mu\rho} g^\sigma{}_\nu$  is given by the antisymmetric part of  $g^{\mu\rho} g^\sigma{}_\nu$ : anti-symmetrizing it yields

$$\Lambda^\mu{}_\nu = \delta^\mu{}_\nu + \frac{1}{2} \omega_{\rho\sigma} (g^{\mu\rho} g^\sigma{}_\nu - g^{\mu\sigma} g^\rho{}_\nu) = \delta^\mu{}_\nu + i \frac{1}{2} \omega_{\rho\sigma} (M^{\rho\sigma})^\mu{}_\nu$$

confronting with the definition of generators

$$g = \mathbb{I} + i \alpha_a T^a$$

yields that the generators are

$$M^{\rho\sigma} = -i (g^{\mu\rho} g^\sigma{}_\nu - g^{\mu\sigma} g^\rho{}_\nu)$$

so the generic Lorentz transformation can be expressed as

$$\Lambda = \exp \left[ \frac{i}{2} \omega_{\rho\sigma} M^{\rho\sigma} \right]$$

and its commutation relations are

$$[M^{\lambda\tau}, M^{\rho\sigma}] = -i (g^{\lambda\sigma} M^{\tau\rho} + g^{\tau\rho} M^{\lambda\sigma} - g^{\lambda\rho} M^{\tau\sigma} - g^{\tau\sigma} M^{\lambda\rho})$$

To make computations easier, the 6 generators are split into 3+3 easier operators: since  $SO(3)$  is a subgroup of  $SO(1,3)$ , there is a way to express half of the Lorentz Group generators in terms of those of  $SO(3)$ :

$$J_i = \frac{1}{2} \varepsilon_{ijk} M^{jk}$$

which obey the  $SO(3)$  algebra

$$[J_i, J_j] = i \varepsilon_{ij}{}^k J_k$$

and generate rotations; the boosts are generated by

$$K^i = M^{0i}$$

which obey the commutation relation

$$[K^i, K^j] = -i \varepsilon^{ij}{}_k J^k$$

which means that the algebra of the  $K$ 's is **not** closed, whereas the algebra of the  $J$ 's is: this is due to the fact that 3D rotations form a subgroup of  $SO(1,3)$ , while boosts do not (it is easy to check that the composition of 2 Cartesian boosts is the composition of a Cartesian boost and a rotation). The mixed commutator yields

$$[J_i, K_j] = i \varepsilon_{ijk} J^k$$

so the set of all  $J$ 's and all  $K$ 's is the set of the generators of  $SO(1,3)$ .

## 5 Lesson 5, 01/10

### 5.1 More on the Lorentz group

The generic Lorentz transformation can also be written in terms of  $J, K$ :

$$\Lambda = \exp [i\vartheta_i J^i + i\eta_i K^i]$$

where  $\vartheta_i$  are the Euler angles (parameters of the rotation) and  $\eta_i$  are the rapidities (parameters of the boost).

With some work, it is possible to prove that

$$\frac{1}{2} M_{\mu\nu} M^{\mu\nu} = J_i J^i - K_i K^i = \mathbf{J}^2 - \mathbf{K}^2$$

and

$$-\frac{1}{4} \varepsilon^{\mu\nu\rho\sigma} M_{\mu\nu} M_{\rho\sigma} = J_i K^i = \mathbf{J} \cdot \mathbf{K}$$

are Casimir operators for the representation: there exist only 2 linear independent Casimir operators and each and any linear combination of these two is valid, but these two are important because they are, respectively, symmetric and antisymmetric (how can they be?? They're scalar!).

**Theorem:** The algebra of the Lorentz Group can be factorized as the direct product of two  $SU(2)$  algebras

$$\mathfrak{so}(1, 3) = \mathfrak{su}(2) \otimes \mathfrak{su}(2)$$

This suggests to find new generators that are suited to this decomposition

$$\begin{aligned} \mathbf{N} &= \frac{1}{2} (\mathbf{J} + i\mathbf{K}) \\ \mathbf{N}^\dagger &= \frac{1}{2} (\mathbf{J} - i\mathbf{K}) \end{aligned}$$

their algebra is

$$\begin{aligned} [N_i, N_j^\dagger] &= 0 \\ [N_i, N_j] &= i\varepsilon_{ijk} N^k \\ [N_i^\dagger, N_j^\dagger] &= i\varepsilon_{ijk} N^{k\dagger} \end{aligned}$$

**Observation:** These generators are convenient, since the algebra of the  $N$ 's and that of the  $N^\dagger$ 's are closed and disjoint.

The Casimir operators are

$$\begin{aligned} N_i N^i &\longrightarrow n(n+1) \\ &\text{eigenvalues} \\ N_i^\dagger N^{i\dagger} &\longrightarrow m(m+1), \end{aligned}$$

so the irreducible representations of the group will be labeled by the pair  $(n, m)$ . Since they obey  $SU(2)$  algebra, it is clear that  $n, m$  will have integer or semi-integer values and that the eigenvalues of  $N_3, N_3^\dagger, s$  and  $s^\dagger$ , will respect

$$-n < s < n \quad -m < s^\dagger < m.$$

To each pair  $n, m$  corresponds a name for the field

| $(n, m)$                   | Field name              | Dim | Example          |
|----------------------------|-------------------------|-----|------------------|
| $(0, 0)$                   | Scalar $\varphi$        | 1   | Higgs            |
| $(1/2, 0)$                 | Spinor $\psi_L$         | 2   | Weyl Spinor      |
| $(0, 1/2)$                 | Spinor $\psi_R$         | 2   | Weyl Spinor      |
| $(1/2, 0) \oplus (0, 1/2)$ | Dirac Spinor            | 4   |                  |
| $(1/2, 1/2)$               | 4-vector field (spin 1) | 4   | $A^\mu$ ELM      |
| $(1, 0) \oplus (0, 1)$     | Rank 2 antisymm tensor  | 6   | $F^{\mu\nu}$ ELM |

## 5.2 Poincaré Group

The QFT must also be invariant over spacetime *translations*: the Poincaré group includes Lorentz transformations *and* translations

$$x^\mu \rightarrow x'^\mu := \Lambda^\mu_\nu x^\nu + a^\mu$$

the group parameters are  $6 + 4 = 10$ . The generators of the 4-translations, similarly to QM, are the 4-momenta

$$P^\mu = (H, P^i);$$

moreover, in QM1 the time evolution operator was found to be

$$U = e^{-iH}$$

which, in hindsight, is an exponential map, in which  $H$  is the generator and  $t$  is the parameter!

There will be some problems with the fact that the translation group is not compact.

Finding the commutation relations for the algebra is not trivial, since rotations and translations do not commute, even in Minkowski space. Consider two successive Poincaré transformations (why????):

$$x^\mu \rightarrow (\Lambda_{(1)})^\mu_\nu x^\nu + a_1^\mu \rightarrow (\Lambda_{(2)})^\mu_\rho (\Lambda_{(1)})^\rho_\nu x^\nu + (\Lambda_{(2)})^\mu_\rho a_1^\rho + a_2^\mu$$

The commutation relations are

$$[P^\mu, P^\nu] = 0$$

$$[M_{\mu\nu}, P_\rho] = -ig_{\rho\sigma}P_\nu + ig_{\nu\rho}P_\mu$$

They are clearer when time and space indexes are expressed separately

$$[P^i, P^j] = 0$$

$$[P^i, H] = 0$$

$$[K^i, P^j] = i\delta^{ij}H$$

$$[J^i, J^j] = i\varepsilon^{ij}_k J^k$$

$$[K^i, H] = iP^i$$

$$[J^i, P^j] = i\varepsilon^{ij}_k P^k$$

$$[J^i, H] = 0$$

$$[K^i, K^j] = i\varepsilon^{ij}_k J^k$$

$$[J^i, K^j] = i\varepsilon^{ij}_k K^k$$

REORDER

Next, the Casimir operators: a natural choice is

$$P^2 = P_\mu P^\mu$$

a not so natural choice is the *Pauli-Lubanski vector* squared:

$$W^\mu = \varepsilon^{\mu\nu\rho\sigma} P_\nu M_{\rho\sigma} \rightarrow W^2 = W_\mu W^\mu.$$

A nice relation is

$$P_\mu W^\mu = 0$$

for antisymmetry reasons.